Did the Structure Emerge, or Was It Prompted? A Structure-Provenance Framework for Synthetic Organizations

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Abstract

Large language model (LLM) agents are increasingly used to study how organizations coordinate, specialize, and develop authority. But when a division of labor or a communication hierarchy appears in a simulation, did it develop through the agents' actions, or was it built into the design? Assigned roles, interaction rules, and even arbitrary labels can produce patterns that resemble organizing.

We introduce structure-provenance analysis to distinguish observed organizational differences from evidence about their origins. It measures whether differences persist beyond appropriate baselines, tests their sensitivity to design changes, and uses controlled interventions to examine how they arise. We evaluate the approach with rule-based agents whose generating mechanisms are known, apply its measures to interaction data from ants, honeybees, workplaces, and online teams, and study LLM-agent organizations.

The results show why these distinctions matter. Initially identical agents complete interdependent work without detectable stable specialization. Assigning specialties produces persistent differentiation but reduces task completion under the tested conditions. An apparent communication hub disappears when arbitrary agent labels are balanced. Coordination, stable roles, and network centrality are therefore distinct outcomes that require different explanations.

Our open repository provides tested measures, null models, complete simulation records, a two-minute guided tour, a one-command report, and executable experimental controls. These resources support reproducible assessments of organizational differentiation, robustness checks, and experimentally grounded comparisons of competing explanations, while allowing the framework to be adapted to other synthetic organizations.

1. The problem of attributing organizational structure

Studies of interacting LLM agents now report network formation, labor specialization, and authority stratification (Papachristou & Yuan 2025; Takata et al. 2024; Ji et al. 2026). Such simulations allow researchers to manipulate aspects of organizing that are difficult to separate in existing groups. But a recurrent difference between members can arise either from their decisions and interactions or from the researcher's specification of roles, routing, exposure, and labels. Observing the difference does not identify its source.

The distinction matters for organizational theory. Completing interdependent tasks is not the same as developing a stable division of labor; a central position in a communication network does not, by itself, establish emergent authority. Evidence about organizational outcomes cannot substitute for evidence about the processes that generated them.

We develop structure-provenance analysis to evaluate such claims. It separates three questions often treated together: Is there durable differentiation, does it depend on arbitrary design features, and what produces it? We validate its measures on rule-based systems with known mechanisms and on interaction records from insects and human groups, then use it to reassess LLM-agent organizations.

2. Structure-provenance analysis

The framework answers three questions, each with a different kind of evidence.

- **Is there structure?** Persistent differentiation above an appropriate baseline. One function takes any activity log and returns dimension-specific persistence measures: repeatability, the share of between-member variance that is stable across windows (Nakagawa & Schielzeth 2010), for how much each member does, which kind of task and who becomes a hub, and partner overlap across windows for with whom. Each is compared with a suitable null: member labels permuted within windows for repeatability, random partner sets for overlap. Scoring work relative to each member's opportunities mitigates unequal task exposure; exposure-preserving and dependence-aware baselines are Stage 2 extensions (D1).
- **Is it robust?** Survival under specified design perturbations: changes to features that should not matter, such as labels and the order of items and recipients.
- **Where did it come from?** Evidence from interventions. Differences can be prescribed (written into prompts), design-induced (order, routing, exposure or label cues) or endogenous (acquired through members' own histories, feedback or interaction), and these sources can coexist. Interventions such as wiping histories, cutting communication, reassigning roles and shuffling labels test competing explanations, subject to the factors each changes; observational logs alone cannot answer this question.

The framework returns a report, not a verdict. For the LLM comparisons in section 3, primary outcomes and decision rules were fixed in a written analysis plan before the new replication results were inspected; post-plan runs alone support the same conclusions. The repository (anonymized for review: anonymous.4open.science/r/synthetic-org-benchmark-45E2) holds the diagnostics, controls, every reported run and a two-minute guided tour that runs without API keys.

What is new. Existing studies report network formation, specialization, and authority among LLM agents (Papachristou & Yuan 2025; Ji et al. 2026; Takata et al. 2024); methodological work addresses experimental design, reproducibility, and validation of LLM-based simulations (Tranchero et al. 2026; Hu & Shen 2026). We address a narrower inferential problem: whether persistent organizational differences can be attributed to agents' histories and interactions rather than to prescribed roles, opportunity structures, routing rules, or incidental cues. The framework combines persistence measurement, null models, design perturbations, and controlled interventions with known-answer validation.

3. Evidence

The three kinds of evidence establish breadth of measurement, identification of known mechanisms, and consequences for reading LLM-agent organizations; the technical appendix reports every run.

A. Measurement across real groups. The same function, unchanged, detects persistent member-level differences in biologically and socially distinct groups, each against its own baseline (Figure 1A): genetically identical ants with no queen and no assigned roles (0.68 against 0.13; Alciatore et al. 2021), office workers at two sites (0.34 and 0.28 against 0.16 and 0.14; Génois et al. 2015; Génois & Barrat 2018) and strangers matched into online game teams (0.51 against 0.15; Cui, Chung & Hanson-Holtry 2017). Preliminary analyses of ants with a queen (Mersch, Crespi & Keller 2013) and honeybees (Gernat et al. 2018) agree. These cases show that the measures detect real-world differentiation. They cannot establish its generating mechanism, because differences in ants or offices cannot be wiped or reassigned. Applied to a published simulation of agents with self-written memory (Takata et al. 2024), the framework finds persistent differences that reach chance when every agent hears every other.

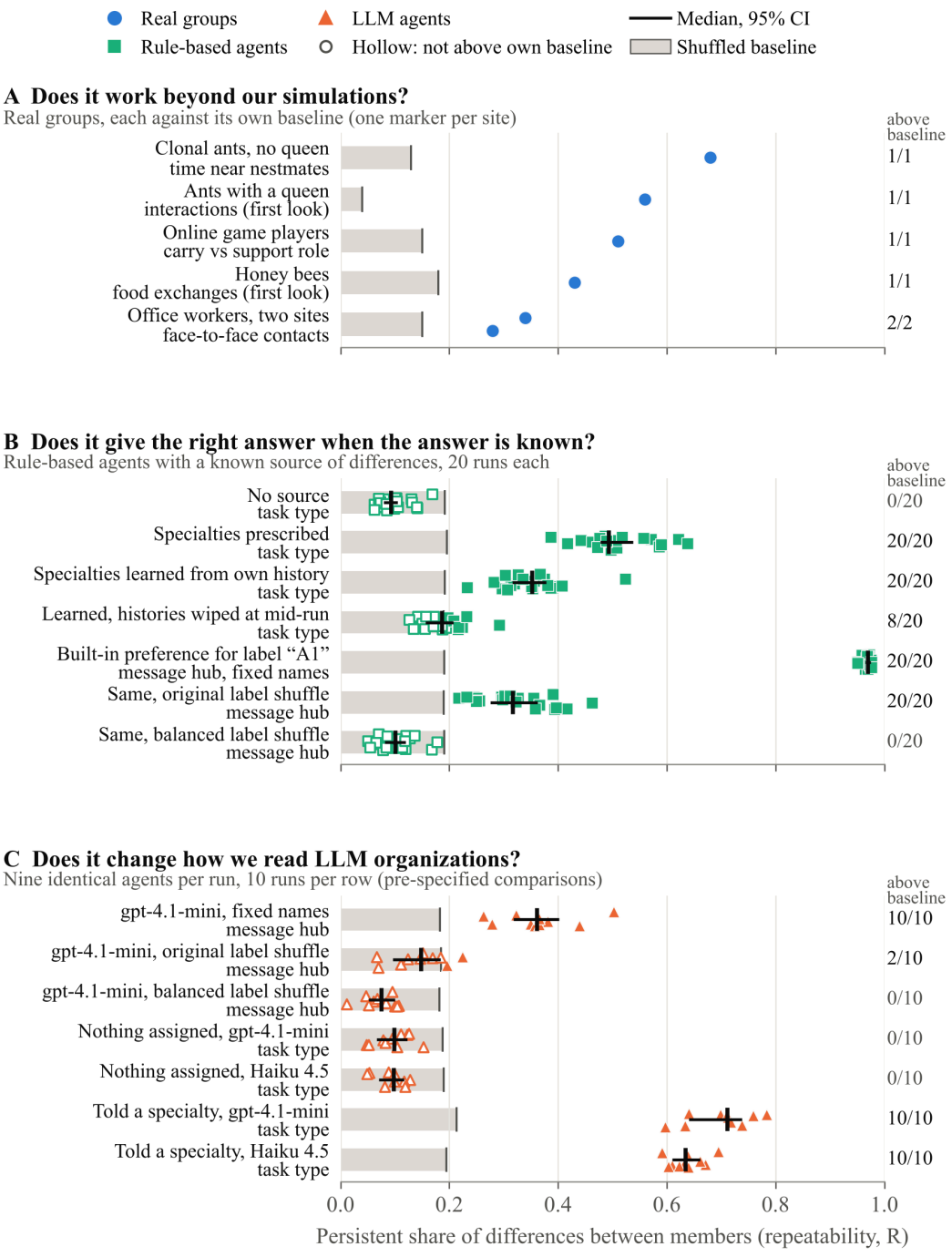


Figure 1. Markers are runs or sites, filled when above their own shuffled baseline (95th percentile of a label-permutation null) and hollow when not. Grey bars run to each row's median baseline; black marks show the median with a bootstrap 95% interval; the right column counts runs above baseline. (A) Real groups. (B) Rule-based agents with known sources of differentiation; for wiped histories, attribution rests on continuity across the wipe, which returns to chance (3B), while specialization can re-form afterwards. (C) Plan-specified comparisons of LLM organizations. Rows record different behaviors, so each is read against its own baseline.

B. Identification under known mechanisms. Rule-based agents in the same environment follow a response-threshold rule, the mechanism that turns identical social insects into specialists (Theraulaz, Bonabeau & Deneubourg 1998). We vary whether specialization is assigned or learned, histories kept, communication allowed and labels shuffled (20 runs each; Figure 1B). The framework detects no persistent task-type differentiation in the no-source condition and detects prescribed and history-dependent specialization in every run. A history wipe separates them: wiping histories halfway leaves prescribed roles intact but returns history-dependent continuity to chance (a drop of 0.19, 95% interval 0.10–0.28). Cutting communication reduces specialization by 0.09 (0.06–0.13) without removing it; because communication also changes how much work agents see, D2 separates interaction from exposure. The test also exposed a flaw in our own stress test: a built-in preference for the label "A1" survives our original label shuffle, because one agent is no one's "A1", while a balanced shuffle removes it in all 20 runs.

C. Coordination without stable specialization. Nine identical LLM agents shared a queue in which each item is revealed to one agent and needs two different workers, so completing it requires telling a colleague. They completed more items than rule-based agents (medians 158 and 118 against 83). Yet in the plan-specified replications no run on either model showed task-type persistence above its baseline (0 of 20; medians 0.10 against baselines of 0.19; 1 of 20 under an alternative calibration, within chance), whereas every run with agents told a specialty did (20 of 20; medians 0.71 and 0.63). The assigned-role conditions, which also constrained which items agents took, completed fewer items: 69% fewer on gpt-4.1-mini, 30% fewer on Haiku 4.5 and 39% fewer among rule-based agents.

D. A hub that followed labels. With fixed names, gpt-4.1-mini agents formed a persistent communication hub in all 10 runs of the plan-specified comparison (median 0.36 against a baseline of 0.18), in eight of them around the agent named "A1". Under our original label shuffle, 2 of 10 runs stayed above baseline; under the balanced shuffle validated in 3B, none did (median 0.08). The label preference remained (88 of 90 agents sent most to whoever they saw as "A1"), so D1 adds a label-aware partner baseline. All four plan-specified contrasts in 3C and 3D hold at $p < 0.001$ (Holm-corrected), also on post-plan runs alone, and at $p < 0.01$ paired by seed.

4. What structure provenance reveals about organizing

Puranam, Alexy and Reitzig (2014) distinguish four problems of organizing: task division, task allocation, reward provision and information provision. Our environment fixes task requirements and collective rewards; agents determine how to allocate work and exchange information. Across two LLM models, initially identical agents completed joint tasks, yet no run showed task-type persistence above its baseline (0 of 20). Assigning specialties increased differentiation but reduced throughput, although the intervention also constrained how work was allocated. The absence of detectable stable specialization is therefore not evidence of deficient coordination: under these conditions, initially equivalent agents completed interdependent tasks without detectable durable task-specific differentiation.

The framework also changes what can be inferred from apparent organizational structure. A persistent communication hub under fixed labels disappeared under balanced relabeling, so observed centrality need not result from an endogenous organizing process. Conversely, persistent differentiation does not by itself reveal whether memory, feedback, interaction or unequal opportunity produced it; Stage 2 (D2) makes that question experimentally tractable.

5. Stage 2: three deliverables

D1. The analysis package. We will freeze the four measures and publish a documented log format. The baselines will add one that preserves each member's exposure, one robust to dependence over time and a label-aware baseline for partner choice; the balanced label shuffle, already validated, replaces the original.

Known-answer tests and controls ship with a worked example and a one-command report. *Success:* an independent researcher runs the package on a compatible log and reproduces our documented results.

D2. A controlled test of endogenous differentiation. We will test whether initially equivalent LLM agents develop persistent task specialization through their own histories, and whether communication adds to it. A factorial experiment varies persistent memory, outcome feedback and access to colleagues' outcomes, holding capabilities, instructions and task opportunities equal. Work passing, which changes opportunities, is analyzed separately, and a condition with hidden comparative advantage separates generating differences from discovering them. Each cell gets a preregistered number of runs on a current proprietary and a current open-weights model. Two Stage 1 contrasts, identical versus assigned-specialty agents and fixed versus balanced labels, are repeated on these models as reference tests and reported by model, without requiring their signs to match. Model versions, conditions, outcomes and interpretation criteria are preregistered before results are examined. *Success:* a clear answer either way: whether memory and feedback suffice under the tested conditions, or a documented boundary condition.

D3. Independent validation and adoption. We will apply the framework to at least two externally developed simulations (one is done) and report how it changes their interpretation. We will test how the reference values respond to window length, exposure and group size, and publish a one-page reporting standard. It serves researchers before they claim emergence, reviewers, and developers comparing agent architectures. *Success:* the framework proves useful beyond the environment that motivated it.

Stage 1's LLM runs cost about $23; D2 costs a few hundred dollars.

6. Risks and safeguards

- **Exchangeability.** Shuffled baselines assume members are exchangeable within a window. Scoring work relative to each member's opportunities mitigates unequal task exposure but does not remove every exposure-related violation; D1 extends the null models to exposure, partner structure and dependence over time.
- **Model advancement and generalizability.** The Stage 1 findings establish how two specified models organize under controlled conditions, not a general limit on LLM capabilities. Later models may develop stable specialization without assigned roles or respond less to incidental cues. Either outcome remains informative: the framework tests whether differentiation persists and what produces it, rather than predicting a particular organizational pattern.
- **Reproducibility and data use.** Every model reply, version and setting is logged. A clean-environment check reproduced all synthetic and stored LLM results from the public repository; reference-group values need the original data deposits. External data are used under their licenses and published pseudonymous identifiers; no new human-subjects data are collected.

7. From observed structure to its provenance

Synthetic organizations let researchers separate experimentally the structure specified at initialization from structure that develops through agents' behavior and interaction, and structure-provenance analysis makes that separation measurable. Stage 1 shows that coordination, persistent specialization and apparent centrality are distinct organizational outcomes that require different evidence to explain. Stage 2 asks the question this makes testable: when, and through which mechanisms, do initially equivalent members develop durable roles of their own?

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